

Assignment 7 : Microwatt Overview

1. What is Microwatt, and which ISA does it implement? Name the organization that develops it and the HDL it is written in.

Microwatt is a soft-core Processor developed by the IBM. The microwatt implements the Power ISA. The development is done by OpenPOWER Foundation. The logic is written in VHDL and can be synthesized using the FPGA for the Prototyping and stuff.

2. List any four architectural features of the Microwatt ISA (e.g., register count, addressing width, endianness support).

The Architectural features of the Microwatt ISA are :

- 32 General Purpose Registers.
 - Implements 64bit architecture.
 - Supports both 32-bit and 64-bit instructions.
 - It supports both Little Endian and Big-Endian formats.
3. Draw/describe the 5-stage pipeline of the Microwatt core and briefly explain the function of each stage.

The 5-stage pipeline of the Microwatt core are :

- Fetch
- Decode
- Execute
- Memory
- Writeback

Fetch Stage :

- In general, the CPU first stage access the set of instructions from ICache and it is the first stage of the CPU pipeline.
- Accesses the PC Program Counter for the instruction address.
- If there any ICache misses or Branch mispredictions then the Fetch stage is blocked or backpressure not to continue in fetching instructions and all stuff.
- Drives the PC into ICache to fetch the instructions.

Decode Stage :

- In general, in the decode stage we decode the instructions in the specified format.
- The Instruction is split into fields, operands and type of the instruction also.
- Expands decoding, handles immediate values and control signals.
- These decoded signals are passed to the information to the Execution stages.

Execution Stage :

- IN general, there are separate execution pipelines like ALU Integer, logical, branch and control operations.
- Drives branch results to fetch stage. If there are any mispredictions to halt the fetching and flushing the pipeline.
- Forwards the data to the Writeback stage.

Memory Stage :

- The Load Store Unit for the memory operations.
- The address generated interact with the DCache for reading and writing the contents to the DCache.
- On any address misses they go the MMU block for address translation and stuff. The MMU is address translation from the Effective address to Physical address.

Writeback Stage :

- Final stage that writes results back to the register file.
- Interacts with debug interface and control/status logic.

4. Why does Microwatt not implement branch prediction? What technique does it use instead?

Design is simple and in order pipeline to avoid the complexity and has stalls and stuff. There are no branch predictors instead we have static or simple branching handling.

5. Explain the role of the ICache and DCache modules in the core. What signals do they use to interface with the Wishbone bus?

ICache :

- Fetches instruction from instruction memory or forwards the request to the Wishbone bus.
- Sends the instructions to the decode1 stage.
- The interface signal is iwbm_signals for memory access.

DCache :

- The Memory load or store operations go to the DCache for read or write operations and stuff.
 - Communicates with MMU for address translation.
 - Interfaces with Wishbone (dwbm_signals).
6. What is the purpose of the MMU module in Microwatt, and what happens on a TLB miss?

The purpose of the MMU (Memory Management Unit) is address translation from Virtual address to the Physical address. If there are any DCache misses for the Load or Store operations and stuff then these are transferred to the MMU block. This goes to the Memory since the Microwatt doesn't have L2 or L3 or LLCC (last level Cache).

Signals TLB misses or page faults to control flow.

7. List any five SoC-level peripheral components (outside the core) and describe their function in one line each.

The SoC level peripheral components are :

- UART
 - In general, there are two ports UART0, UART1.
 - Serial Communication. The UART0 is used as Serial monitor for the device to know what is happening and all stuff. It acts as Console prints of the device.
- SPI
 - External Memory device interface is connected to the SPI.
- Core
 - The CPU core block the real CPU pipeline is present or implemented.
 - The CPU executes the main program and all stuff.
- System Controller
 - This block provides the SoC configuration/status registers.
 - Includes SoC version, reset control, clock config etc.
- Wishbone BRAM Wrapper
 - This block provides the Instruction or data memory to the ICache or DCache.

- The core access the BRAM through the wishbone interface to begin the code execution.
- GPIO
 - These are General Purpose Input or Output Pins. The external world sensors or something can be read or written through these pins.
- External Interrupts Controller for Power ISA
 - In general, the external world on some specific events it halts the execution of the ongoing tasks and service this high priority the program or code execution.

8. Explain the role of xics_ics and xics_icp in Microwatt's interrupt handling.

The role of xics_ics and xics_icp are Interrupt signals from the external world and stuff.

9. Describe the power-up and reset sequence of the Microwatt SoC, from Power-Up to SoC Ready for Full Operation.

- The FPGA powers on, and all logic is initialized to default states.
- A reset signal (soc_reset etc) is asserted to ensure all components start cleanly.
- This reset is propagated to all components in the SoC level blocks : core, BRAM, wishbone fabric, peripherals, interrupt controller, DMI etc.

10. Name three configurable generic parameters of the Microwatt core entity and explain what each control.

The configurable generic parameters of the Microwatt core are :

- a. HAS_FPU, HAS_BTC, EX1_BYPASS : Control FPU inclusion, branch target cache and bypass logic.
- b. ICACHE_NUM_LINES, DCACHE_NUM_WAYS etc : Configure the ICache and DCache geometry.
 - i. **ICache** : 4KB to 8KB. This cache holds the Instructions of the program.
 - ii. **DCache** : Optional and sizes are configurable. This cache holds the Data or Memory instructions access.
- c. ALT_RESET_ADDRESS : Useful for the initialization firmware.